



## Contents

|                                       |          |
|---------------------------------------|----------|
| <b>About Zellic</b>                   | <b>3</b> |
| <hr/>                                 |          |
| <b>1. Overview</b>                    | <b>3</b> |
| 1.1. Executive Summary                | 4        |
| 1.2. Goals of the Assessment          | 4        |
| 1.3. Non-goals and Limitations        | 4        |
| 1.4. Results                          | 4        |
| <hr/>                                 |          |
| <b>2. Introduction</b>                | <b>4</b> |
| 2.1. About Stargate V2                | 5        |
| 2.2. Scope                            | 5        |
| 2.3. Project Overview                 | 5        |
| 2.4. Project Timeline                 | 6        |
| <hr/>                                 |          |
| <b>3. Patch Review</b>                | <b>6</b> |
| 3.1. Notable changes                  | 7        |
| 3.2. Notable behaviors and edge cases | 7        |
| <hr/>                                 |          |
| <b>4. Assessment Results</b>          | <b>8</b> |
| 4.1. Disclaimer                       | 9        |

## About Zellic

Zellic is a vulnerability research firm with deep expertise in blockchain security. We specialize in EVM, Move (Aptos and Sui), and Solana as well as Cairo, NEAR, and Cosmos. We review L1s and L2s, cross-chain protocols, wallets and applied cryptography, zero-knowledge circuits, web applications, and more.

Prior to Zellic, we founded the [#1 CTF \(competitive hacking\) team](#) worldwide in 2020, 2021, and 2023. Our engineers bring a rich set of skills and backgrounds, including cryptography, web security, mobile security, low-level exploitation, and finance. Our background in traditional information security and competitive hacking has enabled us to consistently discover hidden vulnerabilities and develop novel security research, earning us the reputation as the go-to security firm for teams whose rate of innovation outpaces the existing security landscape.

For more on Zellic's ongoing security research initiatives, check out our website [zellic.io](https://zellic.io) and follow [@zellic\\_io](#) on Twitter. If you are interested in partnering with Zellic, contact us at [hello@zellic.io](mailto:hello@zellic.io).



## 1. Overview

### 1.1. Executive Summary

Zellic conducted a security patch review for LayerZero from April 13th to April 14th, 2026. During this engagement, Zellic reviewed Stargate V2's code for security vulnerabilities, design issues, and general weaknesses in security posture.

We were asked to review a patch to Stargate V2 that adds the CreditMessagingMintableBurnable contract, which introduces an emergency-only, owner-controlled mechanism to restore or correct credits. In section [3.7](#), we provide an overview of the changes.

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### 1.2. Goals of the Assessment

In a security assessment, goals are framed in terms of questions that we wish to answer. These questions are agreed upon through close communication between Zellic and the client. In this assessment, we sought to answer the following question:

- Were any bugs introduced by adding a new CreditMessagingMintableBurnable contract?
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### 1.3. Non-goals and Limitations

We did not assess the following areas that were outside the scope of this engagement:

- Front-end components
- Infrastructure relating to the project
- Key custody

Due to the time-boxed nature of security assessments in general, there are limitations in the coverage an assessment can provide.

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### 1.4. Results

During our assessment on the scoped Stargate V2 contracts, there were no security vulnerabilities discovered.

## 2. Introduction

### 2.1. About Stargate V2

LayerZero contributed the following description of Stargate V2:

Stargate is a composable cross-chain liquidity transport protocol enabling seamless asset transfers between blockchains. [Its] documentation details its technical architecture, core primitives and integration guidelines for developers.

### 2.2. Scope

The engagement involved a review of the following targets:

#### Stargate V2 Contracts

|            |                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| Type       | Solidity                                                                                                          |
| Platform   | EVM-compatible                                                                                                    |
| Target     | Only changes between 1d37fcc...c2c9c7f                                                                            |
| Repository | <a href="https://github.com/stargate-protocol/stargate-v2">https://github.com/stargate-protocol/stargate-v2</a> ↗ |
| Version    | c2c9c7fb0a3867b590257fea6ee13ad87a24ed56                                                                          |
| Programs   | packages/stg-evm-v2/src/messaging/CreditMessagingMintableBurnable.sol                                             |

### 2.3. Project Overview

Zellic was contracted to perform a security assessment for a total of 0.5 person-weeks. The assessment was conducted by two consultants over the course of two calendar days.

## Contact Information

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The following project manager was associated with the engagement:

**Jacob Goreski**  
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The following consultants were engaged to conduct the assessment:

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**Quentin Lemauf**  
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## 2.4. Project Timeline

The key dates of the engagement are detailed below.

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**April 13, 2026** Start of primary review period

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**April 14, 2026** End of primary review period

## 3. Patch Review

This section documents changes applied to the in-scope code from commit [1d37fcc3](#) to commit [c2c9c7fb](#).

### 3.1. Notable changes

The following were notable changes made to the codebase.

#### CreditMessagingMintableBurnable contract

The `CreditMessagingMintableBurnable` contract extends `CreditMessaging` to introduce two owner-controlled emergency functions: `mintCredits` and `burnCredits`. Both functions require a nonempty `_reason` string and operate directly on the current chain without sending a `LayerZero` message.

The `mintCredits` function applies credits to one or more paths by calling `receiveCredits` locally on the corresponding Stargate implementation for each batch entry. It passes `0` as the source endpoint ID, reflecting that these credits are injected out of band rather than received from a remote chain. A `CreditsMinted` event is emitted with the full input batch upon completion.

The `burnCredits` function deducts credits from one or more paths by calling `sendCredits` locally, forwarding each entry's `minAmount` as `minKept` to the underlying handler. The actually burned amounts are collected and emitted in a `CreditsBurned` event.

### 3.2. Notable behaviors and edge cases

The following section highlights behaviors of the `CreditMessagingMintableBurnable` contract that do not constitute security vulnerabilities but are important operational considerations for owners invoking `mintCredits` and `burnCredits`.

#### The UNLIMITED\_CREDIT path handling

Paths whose credit is set to `UNLIMITED_CREDIT` (`type(uint64).max`) represent OFT paths and are treated specially by `PathLib`. This sentinel value cannot be set or unset by `mintCredits` or `burnCredits`; it is managed exclusively through `setOFTPath`.

In `mintCredits`, `PathLib.increaseCredit` silently returns early when the current credit equals `UNLIMITED_CREDIT`, leaving the path unchanged. Consequently, if `mintCredits` is called with a batch that includes an OFT path, that path's credit will not be updated, yet the `CreditsMinted` event will still be emitted covering the full input batch — including the skipped entry. The emitted event therefore does not accurately reflect which paths were actually modified.

In `mintCredits` (overflow into `UNLIMITED_CREDIT`), if a path's current credit plus the supplied amount equals exactly `type(uint64).max`, `PathLib.increaseCredit` reverts with `Path_UnlimitedCredit`. Because `mintCredits` processes batches in a single transaction, a single entry whose resulting credit would hit `type(uint64).max` causes the entire call to revert.

Owners should verify that no path in a batch is close enough to `type(uint64).max` to overflow into the sentinel value before submitting.

**In `burnCredits`,** `PathLib.tryDecreaseCredit` reverts with `Path_UnlimitedCredit` when the current credit equals `UNLIMITED_CREDIT`. Therefore, unlike `mintCredits`, `burnCredits` will revert — rather than silently skip — if any entry in the batch targets an OFT path.

**In `burnCredits (zero decrease)`,** if `_minKept >= currentCredit`, `tryDecreaseCredit` returns 0 without modifying state. This is by design; the `sendCredits` caller treats a zero return as "nothing to burn" and omits that entry from the returned credits array — `CreditsBurned` only includes entries that burned a positive amount.



## 4. Assessment Results

During our assessment on the scoped Stargate V2 contracts, there were no security vulnerabilities discovered.

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### 4.1. Disclaimer

This assessment does not provide any warranties about finding all possible issues within its scope; in other words, the evaluation results do not guarantee the absence of any subsequent issues. Zellic, of course, also cannot make guarantees about any code added to the project after the version reviewed during our assessment. Furthermore, because a single assessment can never be considered comprehensive, we always recommend multiple independent assessments paired with a bug bounty program.

For each finding, Zellic provides a recommended solution. All code samples in these recommendations are intended to convey how an issue may be resolved (i.e., the idea), but they may not be tested or functional code. These recommendations are not exhaustive, and we encourage our partners to consider them as a starting point for further discussion. We are happy to provide additional guidance and advice as needed.

Finally, the contents of this assessment report are for informational purposes only; do not construe any information in this report as legal, tax, investment, or financial advice. Nothing contained in this report constitutes a solicitation or endorsement of a project by Zellic.